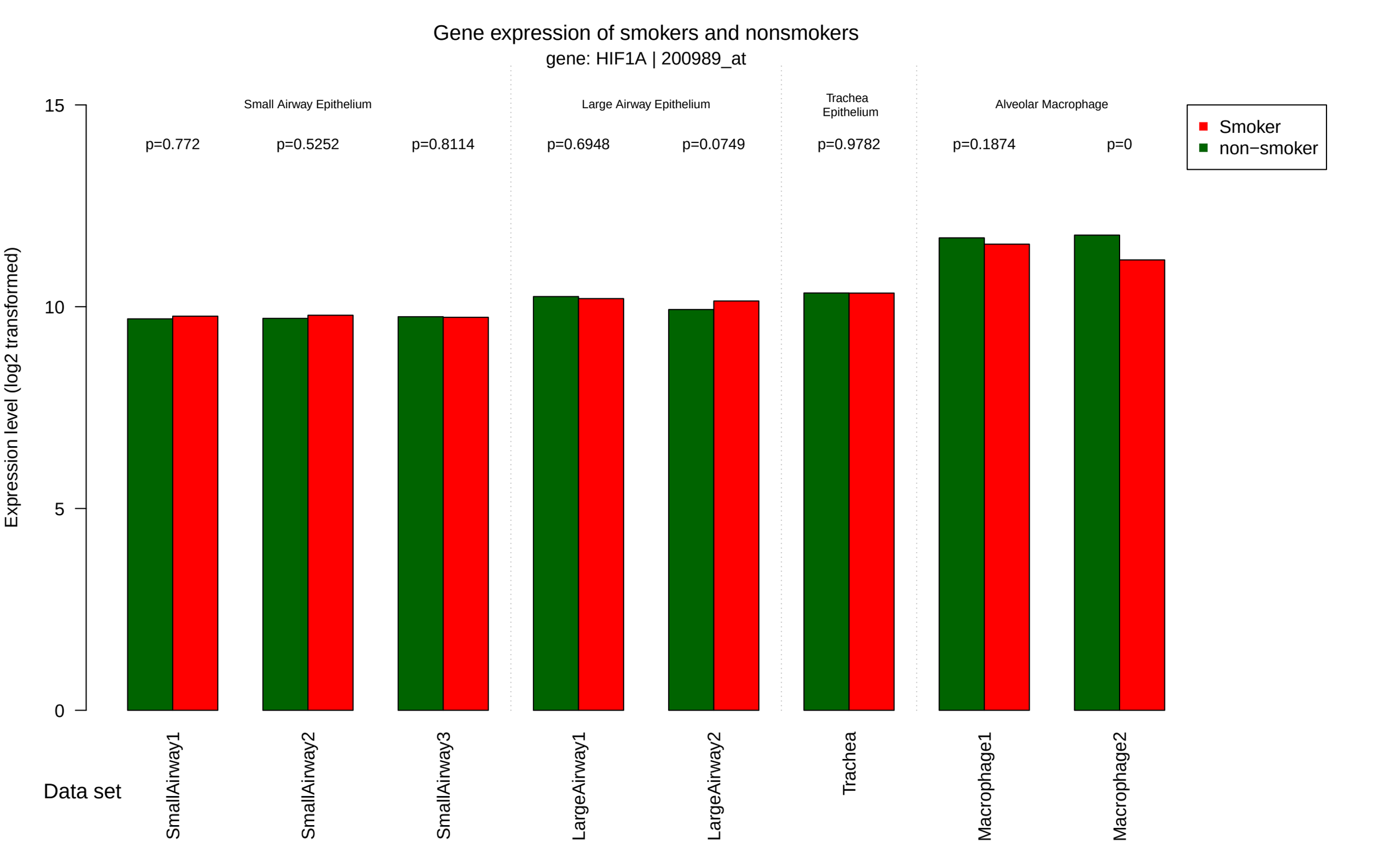
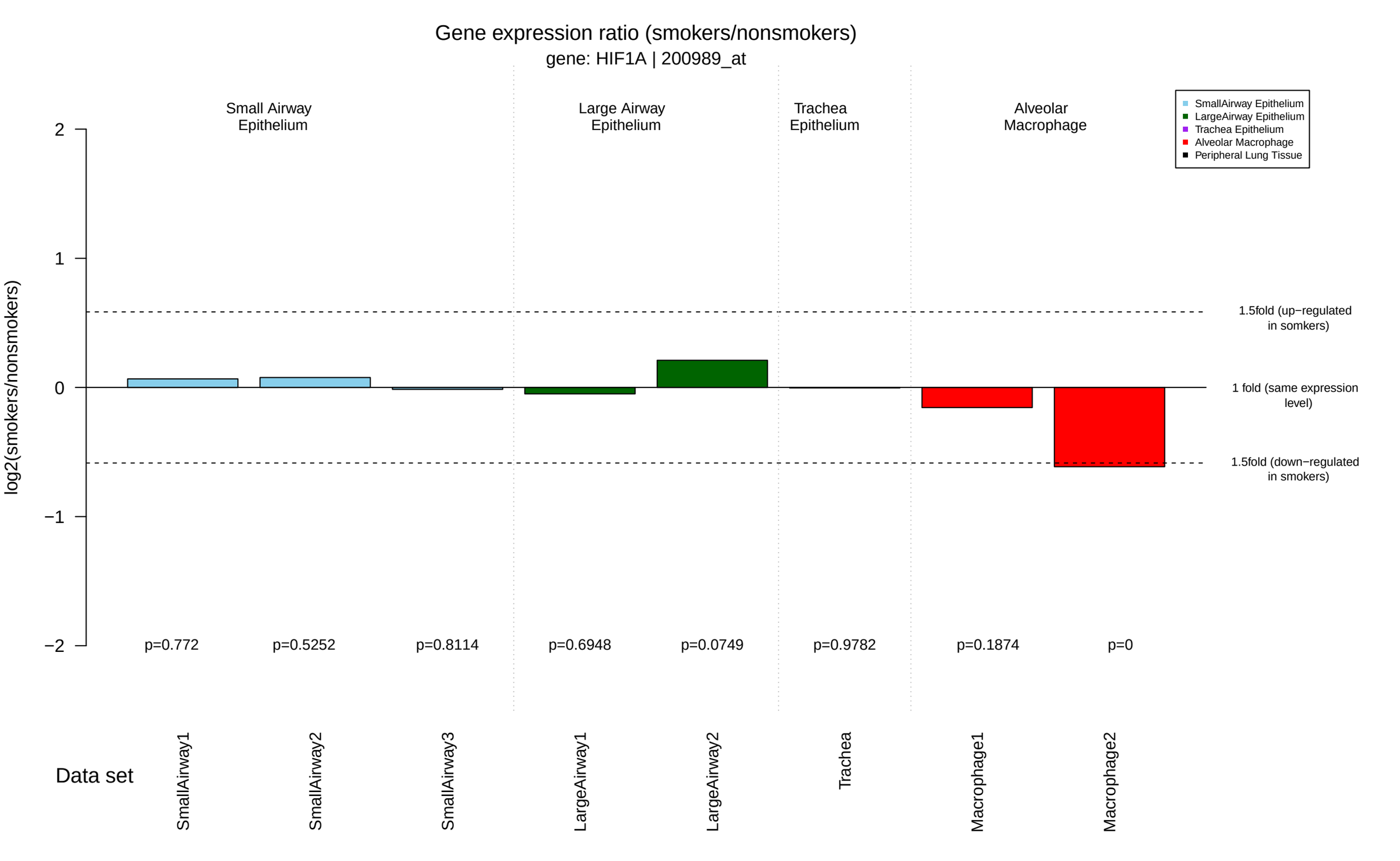
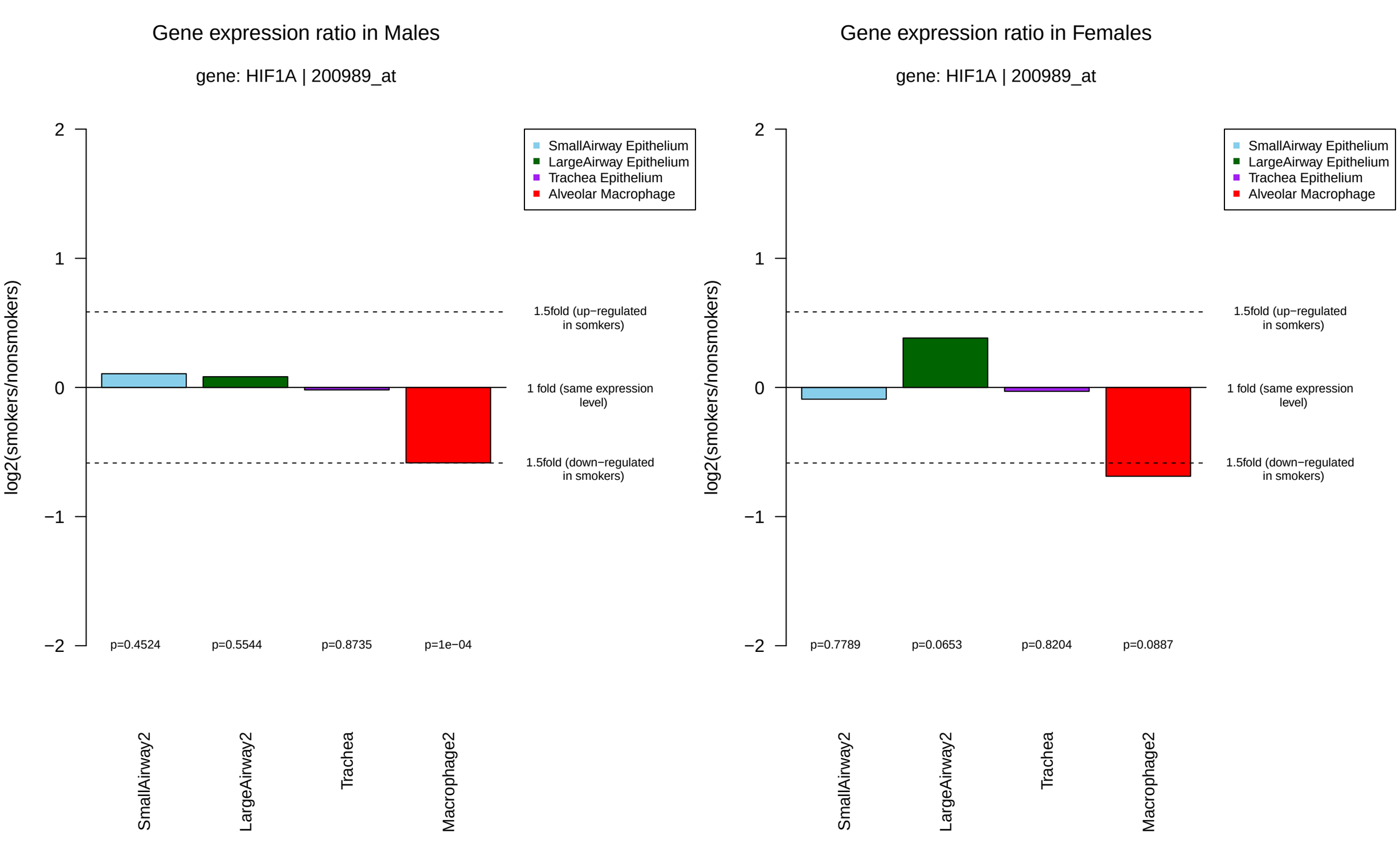
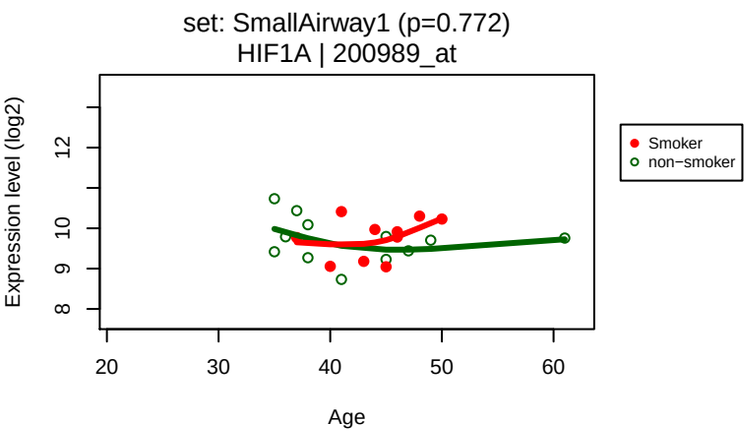
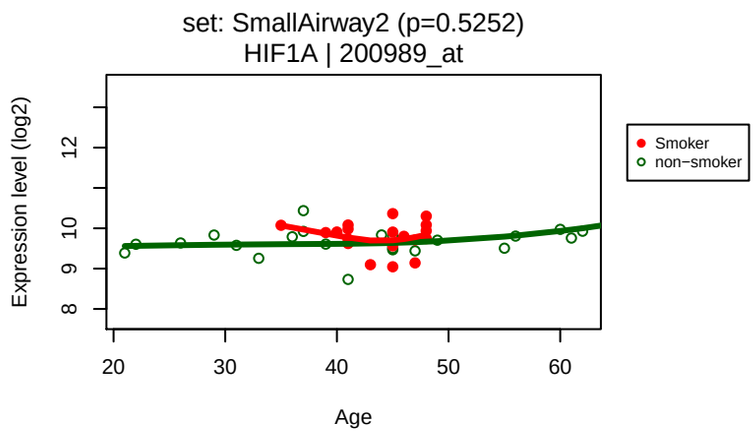
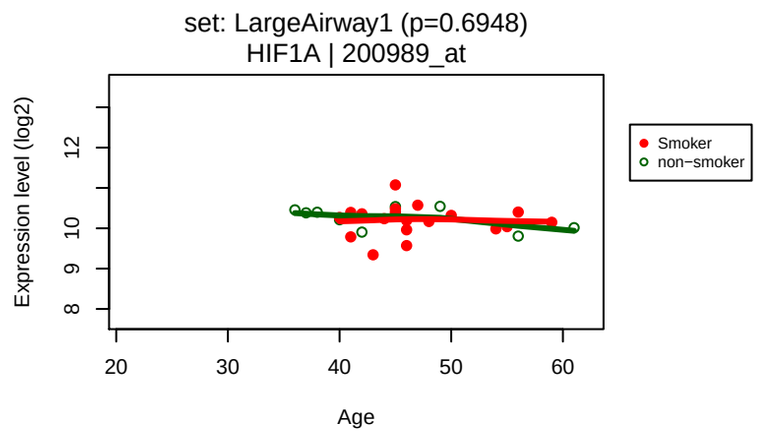

Gene expression of smokers and nonsmokers
gene: HIF1A | 200989_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.772
p=0.5252
p=0.8114
p=0.6948
p=0.0749
p=0.9782
p=0.1874
p=0
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

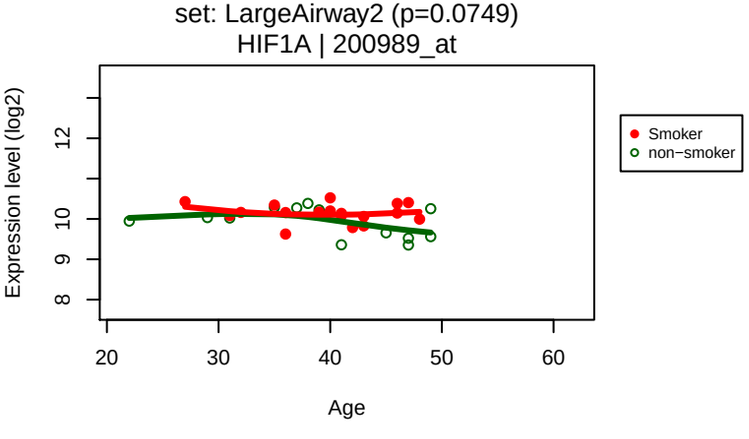

Gene expression ratio (smokers/nonsmokers)
gene: HIF1A | 200989_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.772
p=0.5252
p=0.8114
p=0.6948
p=0.0749
p=0.9782
p=0.1874
p=0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

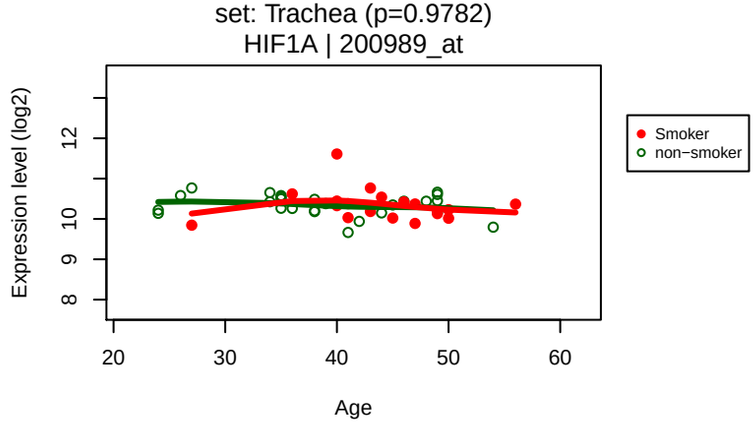

Gene expression ratio in Males
gene: HIF1A | 200989_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4524
p=0.5544
p=0.8735
p=1e−04
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HIF1A | 200989_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7789
p=0.0653
p=0.8204
p=0.0887
SmallAirway2
LargeAirway2
Trachea
Macrophage2

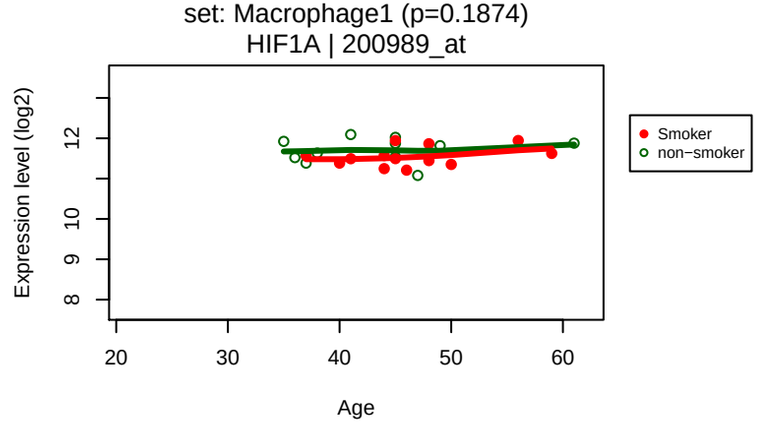

set: SmallAirway1 (p=0.772)
HIF1A | 200989_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker

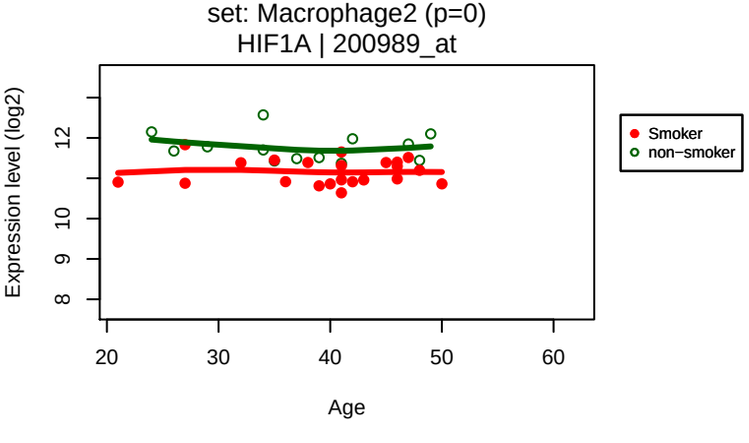

set: SmallAirway2 (p=0.5252)
HIF1A | 200989_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker

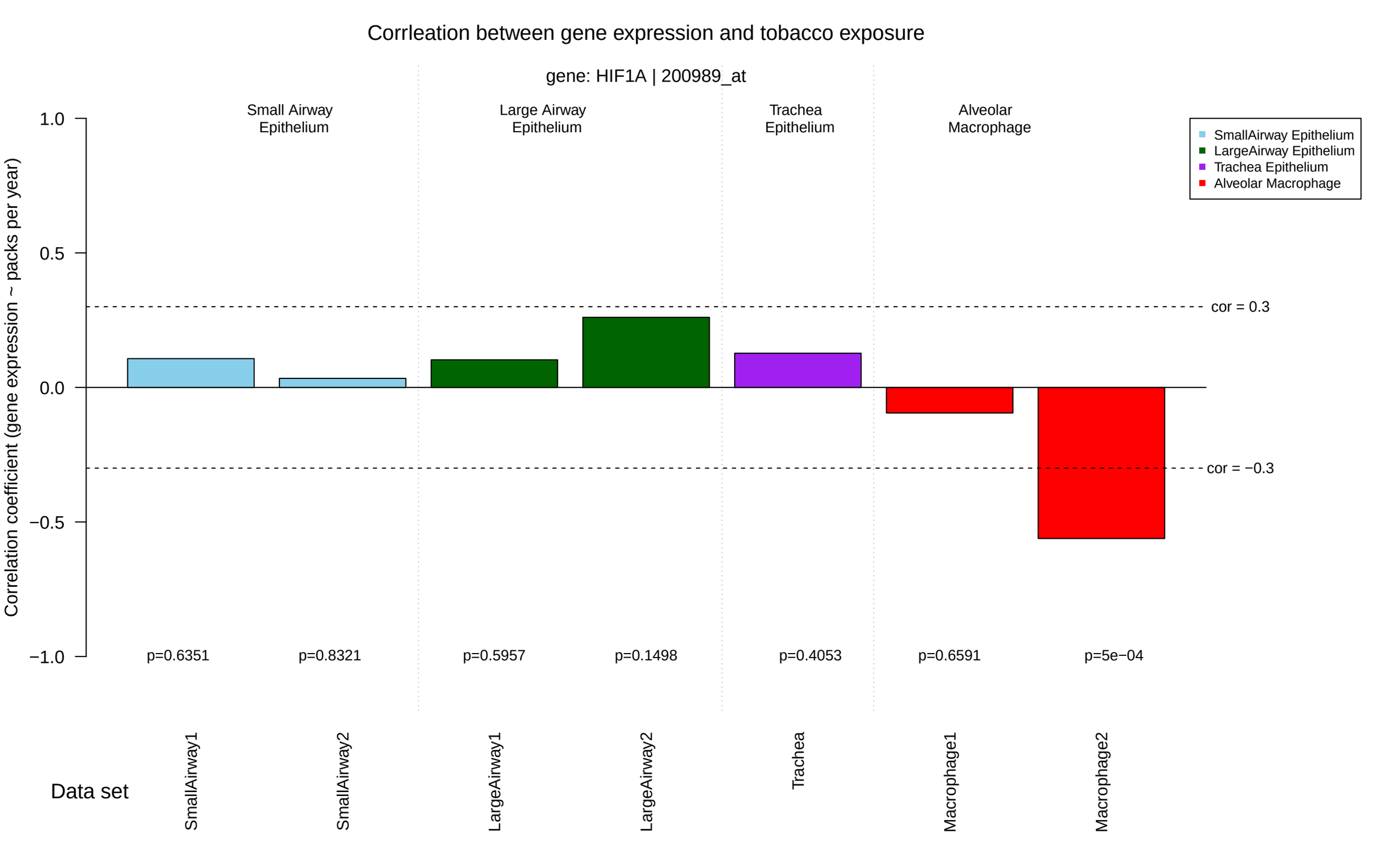

set: LargeAirway1 (p=0.6948)
HIF1A | 200989_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0749)
HIF1A | 200989_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.9782)
HIF1A | 200989_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1874)
HIF1A | 200989_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0)
HIF1A | 200989_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HIF1A | 200989_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6351
p=0.8321
p=0.5957
p=0.1498
p=0.4053
p=0.6591
p=5e−04
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2